

Name: N. Harshith

Course: Computer vision

Reg No: 192325078

Code: JTA0508

AT-1

1. Grey level transformation

a) Enhances image brightness and contrast by changing gray levels.

b) Negative transformation $s = L-1-r$ (for 8 bits)

2) log transformation

a) Expands dark pixel and compress brighten pixels, improving low intensity details.

b) formula $= s = \log(1+r)$

3) Histogram processing

a) Histogram equalization improves image contrast by redistribution intensity volumes.

b) CDF = Cumulative sum of histogram probabilities.

4) Spatial filtering (smoothing)

a) Removes noise & smooths the image

b) New center pixel = sum of 3x3 pixels.

5) Spatial filtering (sharpening)

a) Enhances edges and the details.

b) Output = sum of (mask x corresponding pixels)

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6) Frequency domain filtering

a) low pass removes high frequencies.

b) High pass removes low frequencies.

7) Thresholding

a) Separates object and background using threshold.

b) if $\text{pixel} > T \rightarrow 1$, otherwise $\rightarrow 0$.

8) Edge detection

a) detects sudden intensity changes.

b) Gradient = sum of (Sobel mask $\times$ image pixel)

9) Region based segmentation.

a) starts from a seed and add similar neighbour pixels.

b) Add similar pixel neighbour.

10) Combined enhancement & segmentation.

a) smoothing Remove noise before segmentation.

b) Average first, threshold. $\text{pixel} > T \rightarrow 1$, otherwise $\rightarrow 0$.

11) Power Law Transformation

a) ^{Gamma.} ~~Gamma~~ controls image brightness.

b) $\gamma < 1$ brightness, $\gamma > 1$ darkness.

12) Contrast stretching

a) Improves low contrast images.

b) Linear mapping $S = (r - \min) (\gamma_{\max} - \gamma_{\min}) + 55$

13) Histogram Equalization

a) Improves contrast

b) Uses CDF for mapping.

14. Median filtering

a) Salt and pepper noise contains black & white spikes.

b) Median filter removes impulse noise

15. Gaussian Smoothing

a) Reduce noise

b) Better than the overlapping, because it gives more weight to the center.

c) Produces smooth images.

16. High pass filtering

a) Sharpen the image

b) Highlights edges

17) Frequency domain high pass

a) High frequency represents edges.

b) Removes low frequency

18) Adaptive thresholding

a) Handles uneven illumination.

b) Uses local threshold instead of global.

19) Vertical Sobel

a) Detects vertical edges.

b) More sensitive to vertical intensity changes.

20) Prewitt edge detection

a) Detect edges.

b) simpler than sobel.

c) Apply prewitt kernel

21) Region splitting

a) separates dissimilar region.

b) Based on threshold.

c) threshold = 80 $\rightarrow$ region 1: [40, 45], region (100, 100]

22) Region Merging.

a) Combines similar regions.

b) Merge if differences $\leq$ threshold.

c) Differences $\rightarrow 2 \leq 5 \rightarrow$ merged. $[45, 50, 52, 55]$

23) Edge Linking

a) Connects broken edges.

b) Uses connectivity

c) $[1, 0, 1, 0, 1] \rightarrow [1, 1, 1, 1, 1]$ [linked]

d) Continuous boundary obtained.

24) Low pass filtering

a) Removes noise

b) Smooth images.

c) Retain low frequency only.

25) Laplacian Sharpening

a) Enhances fine details.

c) d) Apply Laplacian

b) Uses second derivative

e) Edges become sharp.

26) Histogram matching

a) Matches one histogram to another

b) Uses CDF mapping

7) Freque

a) Hig

b) R

8) Adap

a)

b)

19) Ver

a)

b)

20)

21)

27) Noise Reduction Comparison

a) Salt and Pepper noise

b) Median is better than overlapping

28) Combined filtering

a) Removes noise than sharpness.

b) Smooth first, sharp next

c) Apply averaging $\rightarrow$ sharpening

d) Cleaner and sharper image

29) Multi threshold segmentation.

a) Divides image into multiple changes.

b) threshold = 50, 100

c) 20 $\rightarrow$ class 1, 60 $\rightarrow$ class 2, 20 $\rightarrow$ class 3 $\rightarrow$ 180

e) Produces three segmentation regions.

30) Boundary detection

a) Extracts object boundaries.

b) Uses edge detection and linking.

c) Detect edges $\rightarrow$ Link connected edges.

e) Continuous object boundaries are obtained.